Experiment 15: Direct Linear Transformation (DLT)

AIM

To perform image transformation using the Direct Linear Transformation (DLT) method.

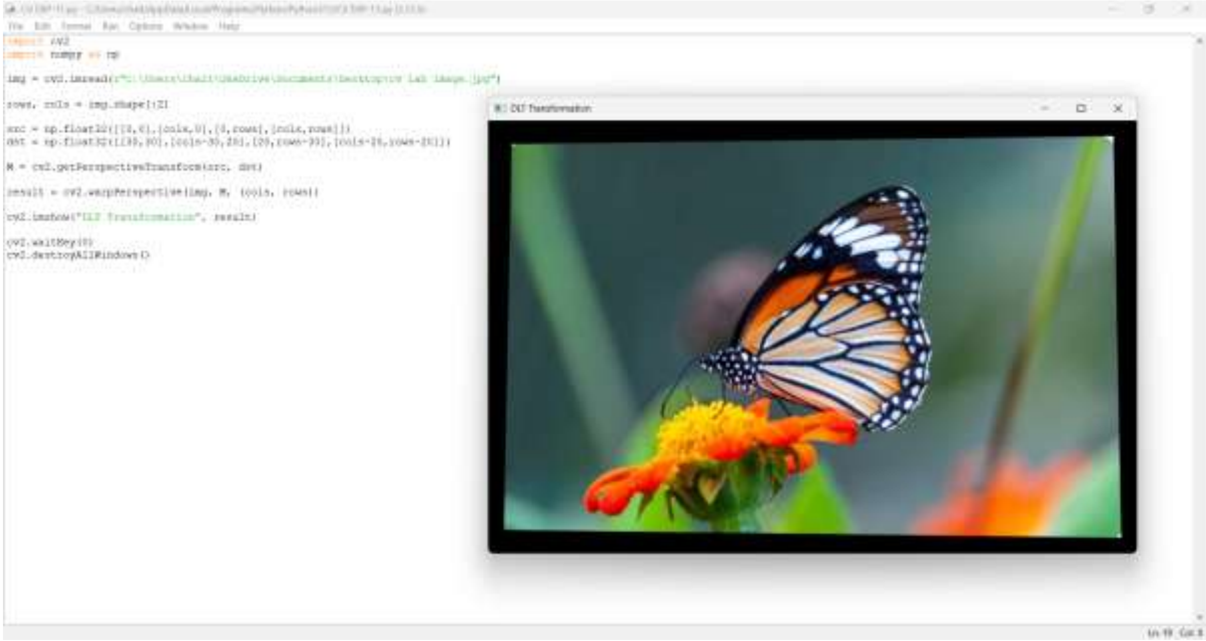
PROCEDURE

1. Read the image.
2. Define source and destination points.
3. Compute the transformation matrix.
4. Warp the image.
5. Display the transformed image.

PROGRAM

```
import cv2
import numpy as np

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

rows, cols = img.shape[:2]

src = np.float32([[0,0],[cols,0],[0,rows],[cols,rows]])
dst = np.float32([[30,30],[cols-30,20],[20,rows-30],[cols-20,rows-20]])

M = cv2.getPerspectiveTransform(src, dst)

result = cv2.warpPerspective(img, M, (cols, rows))

cv2.imshow("DLT Transformation", result)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

OUTPUT:



RESULT:

The image is displayed after Direct Linear Transformation.